1 Convergence

Let $f \in L^1(\mathbb{R})$ and suppose $f_n \to f$ pointwise.

Theorem 1 (Dominated Convergence)**.** *If* $|f_n| \leq g$ *with* $g \in L^1$*, then*

$$\lim_{n\to\infty} \int_{-\infty}^{\infty} f_n(x)\,dx = \int_{-\infty}^{\infty} f(x)\,dx. \tag{1}$$

Proof. Apply Fatou's lemma to $g + f_n$ and to $g - f_n$:

$$\int g + \int f \leq \liminf \int (g + f_n) \tag{2}$$

$$= \int g + \liminf \int f_n. \tag{3}$$

$\square$

2 Assorted formulas

A piecewise definition,

$$f(x) = \begin{cases} x^2 & x \geq 0, \\ -x^2 & x < 0, \end{cases}$$

a matrix,

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix},$$

and a sum with Greek letters: $\sum_{k=1}^{n} \alpha_k \beta_k \leq \gamma$. The complexity is $O(n^2)$ and the radius is $r^{4/3}$.